



A generalization of the Funk–Radon transform

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Mathematik!
TU Chemnitz

- Sphere $\mathbb{S}^2 = \{\xi \in \mathbb{R}^3 : \|\xi\| = 1\}$
- Function $f: \mathbb{S}^2 \rightarrow \mathbb{C}$
- Circles on the sphere are intersections of the sphere with planes:

$$\{\eta \in \mathbb{S}^2 : \langle \xi, \eta \rangle = x\},$$

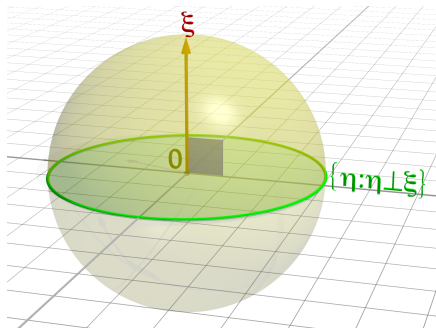
for $\xi \in \mathbb{S}^2$, $x \in [-1, 1]$

- **Spherical Radon transform** (a.k.a. **Funk-Radon transform**)

$$\mathcal{F}: C(\mathbb{S}^2) \rightarrow C(\mathbb{S}^2),$$

$$\mathcal{F}f(\xi) = \int_{\langle \xi, \eta \rangle = 0} f(\eta) d\lambda(\eta)$$

[Funk, 1913]



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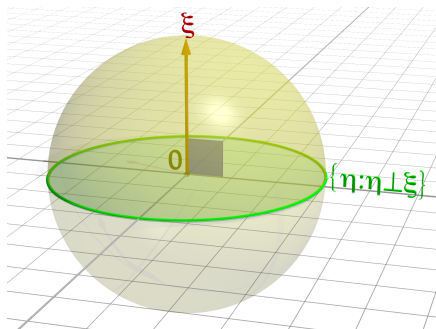
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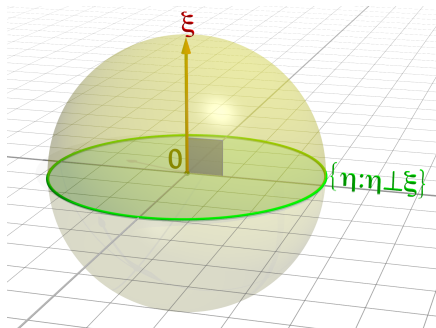
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Generalized Radon transform

[Salman, 2015]

Replace 0 by an arbitrary point

$$\zeta = (0, 0, z)^\top, \quad 0 \leq z < 1$$

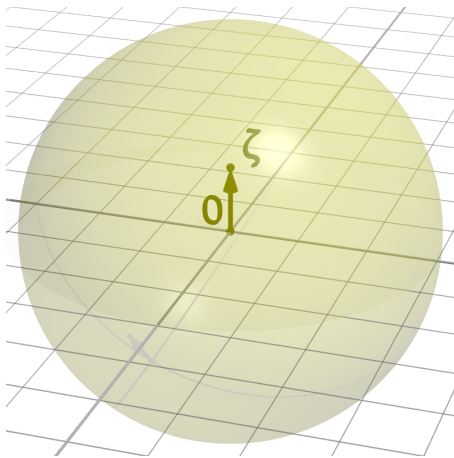
inside the sphere.

Circle through ζ is

$$\{\eta \in S^2 : \langle \xi, \eta \rangle = \underbrace{\langle \xi, \zeta \rangle}_{=z\xi_3}\}.$$

Definition

$$\begin{aligned} \mathcal{U}: C(S^2) &\rightarrow C(S^2), \\ \mathcal{U}f(\xi) &= \int_{\langle \xi, \eta \rangle = z\xi_3} f(\eta) \, d\lambda(\eta) \end{aligned}$$



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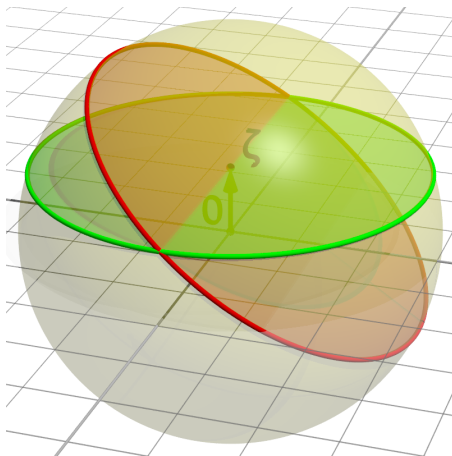
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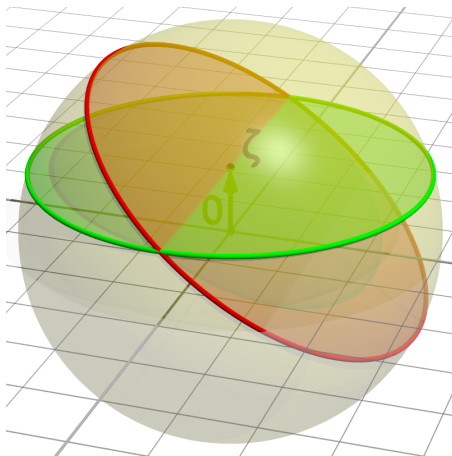
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What is known about the Funk–Radon transform

Theorem

[Strichartz, 1981]

The Funk–Radon transform (spherical Radon transform)

$$\mathcal{F}: L_e^2(\mathbb{S}^2) \rightarrow H_e^{1/2}(\mathbb{S}^2)$$

is continuous and bijective.

Its nullspace consists of all odd functions $f(\xi) = -f(-\xi)$.

Question

What about the generalized Radon transform?

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Question

What about the generalized Radon transform?

From great circles to small circles

Definition

Define the map

$$h: \mathbb{S}^2 \rightarrow \mathbb{S}^2, \quad h = \pi^{-1} \circ \sigma \circ \pi$$

consisting of

1. Stereographic projection $\pi: \mathbb{S}^2 \rightarrow \mathbb{R}^2$
2. Scaling in the plane $\sigma: \mathbb{R}^2 \rightarrow \mathbb{R}^2, \quad x \mapsto \sqrt{\frac{1+z}{1-z}} x$
3. Inverse stereographic projection $\pi^{-1}: \mathbb{R}^2 \rightarrow \mathbb{S}^2$

We show that

h maps great circles to small circles through $\zeta = (0, 0, z)^\top$.

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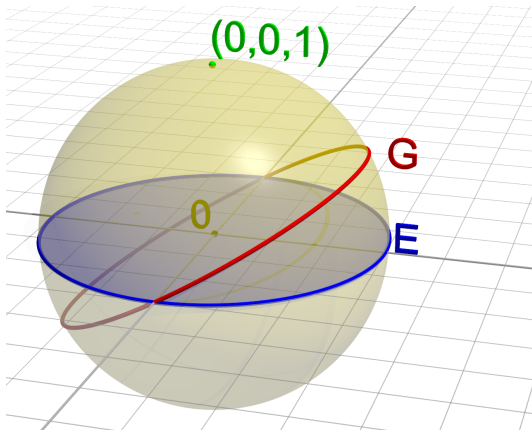
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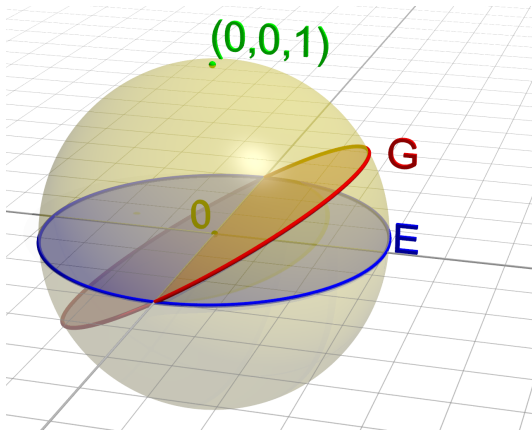
1) Stereographic projection π

- ▶ G ... Great circle of $\mathbb{S}^2$
- ▶ E ... Equator of $\mathbb{S}^2$
- ▶ G intersects E in two antipodal points (or is identical to E)
- ▶ $\pi(E) = E$
- ▶ $\pi(G)$ is a circle or line in $\mathbb{R}^2$ that intersects $\pi(E)$ in two antipodal points



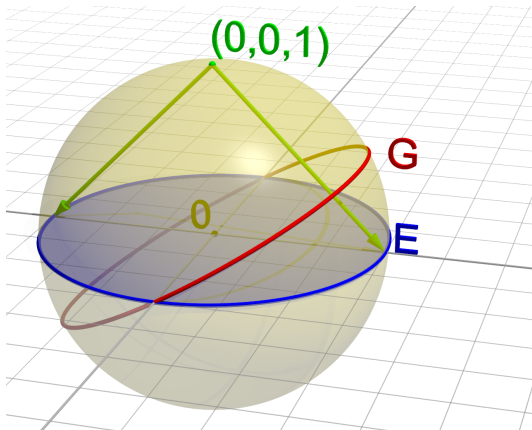
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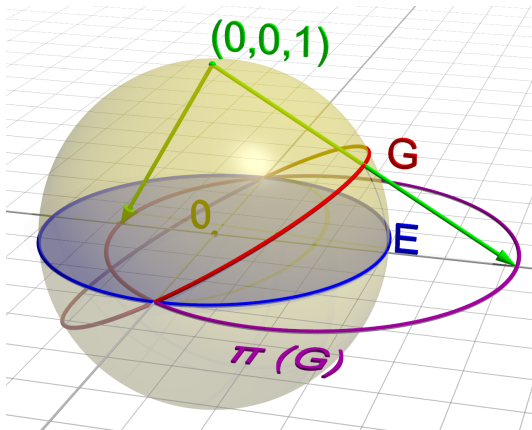
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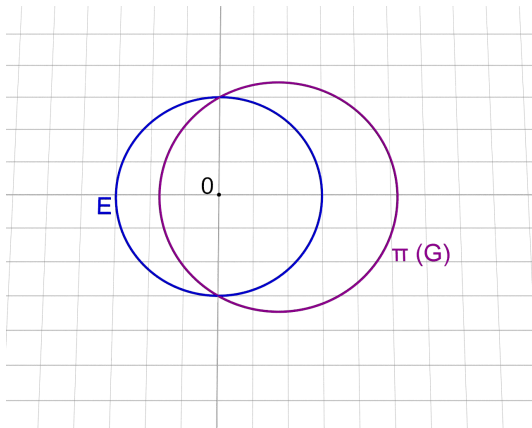
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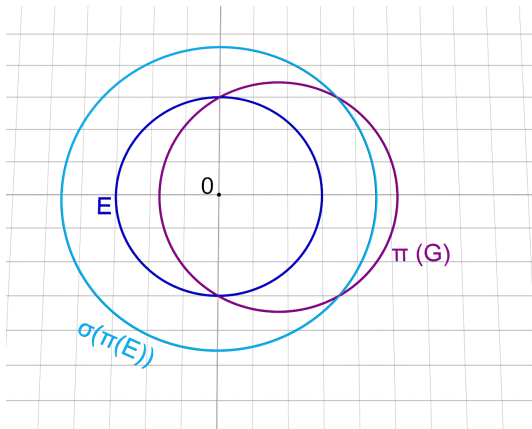
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- Uniform scaling with scale factor $s = \sqrt{\frac{1+z}{1-z}}$
- Unit circle E is mapped to the circle $\sigma(\pi(E))$ with radius s
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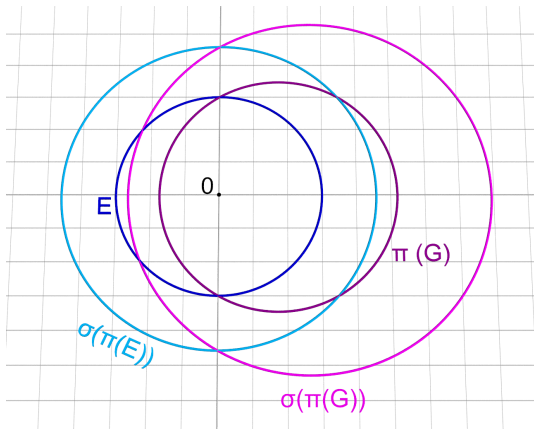
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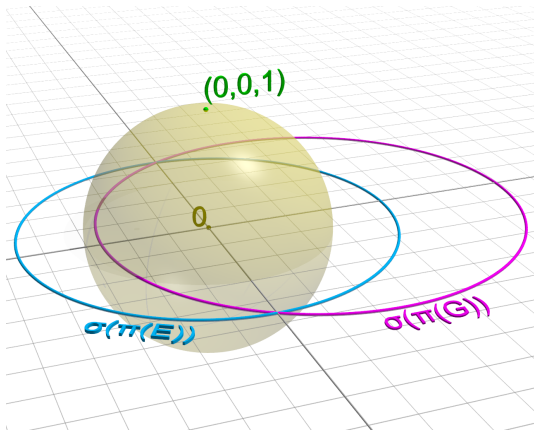
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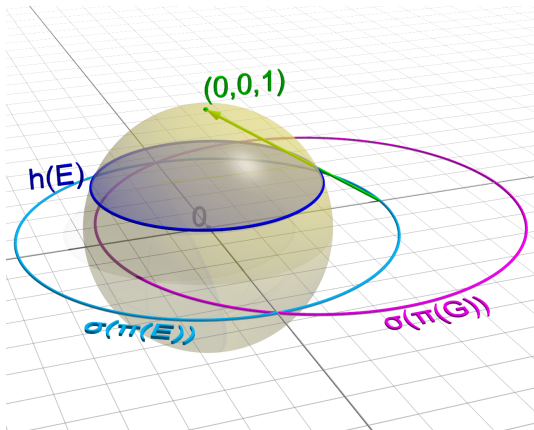
3) Inverse stereographic projection π^{-1}

- ▶ The circle with radius s is mapped to the circle of latitude z ; $h(E)$
- ▶ $h(G) = \pi^{-1}(\sigma(\pi(G)))$ intersects $h(E)$ in two antipodal points
- ▶ $h(G)$ is a small circle through $\zeta = (0, 0, z)^\top$



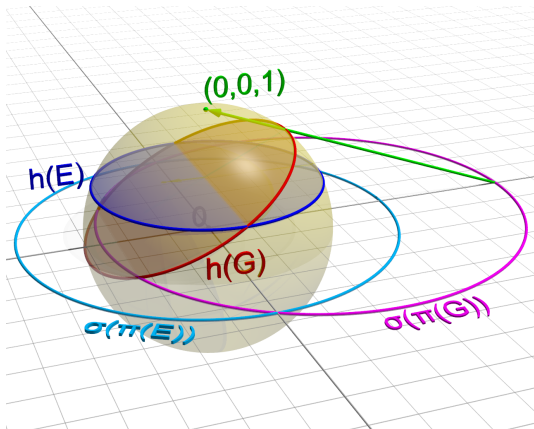
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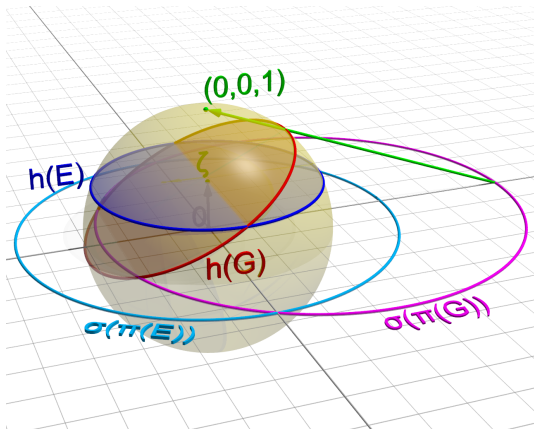
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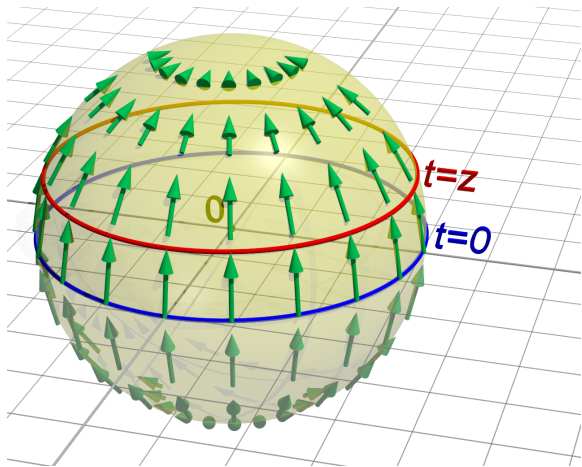


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The resulting map $h: \mathbb{S}^2 \rightarrow \mathbb{S}^2$



$$h(\eta) = \begin{pmatrix} \frac{\sqrt{1-z^2}}{1-z\eta_3} \eta_1 \\ \frac{\sqrt{1-z^2}}{1-z\eta_3} \eta_2 \\ \frac{z+\eta_3}{1-z\eta_3} \end{pmatrix}$$

Theorem

Let $z \in [0, 1)$. The generalized Radon transform $\mathcal{U}$ can be represented with the operators $\mathcal{M}, \mathcal{F}, \mathcal{N}: L^2(\mathbb{S}^2) \rightarrow L^2(\mathbb{S}^2)$ via

$$\mathcal{U} = \mathcal{N}\mathcal{F}\mathcal{M}.$$

These operators are defined for $f \in C(\mathbb{S}^2)$ by

- ▶ $\mathcal{M}f(\xi) = \frac{\sqrt{1-z^2}}{1+z\xi_3} [f \circ h](\xi)$
- ▶ $\mathcal{F}$... Funk–Radon transform
- ▶ $\mathcal{N}f(\xi) = f\left(\frac{1}{\sqrt{1-z^2\xi_3^2}}\left(\xi_1, \xi_2, \sqrt{1-z^2}\xi_3\right)\right)$

Nullspace of $\mathcal{U}$

Theorem

$\mathbf{R}$... Reflection of the sphere about the point $\zeta = (0, 0, z)^\top$

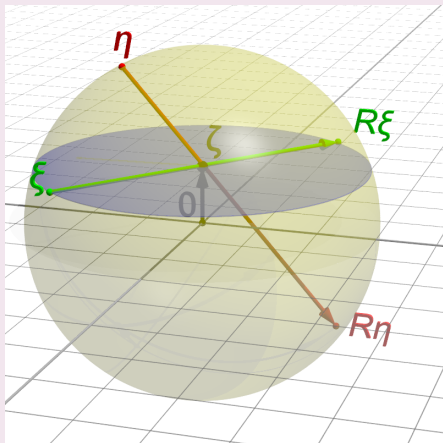
$$f \in L^2(\mathbb{S}^2)$$

We have

$$\mathcal{U}f = 0$$

if and only if for almost all $\eta \in \mathbb{S}^2$

$$f(\eta) = -f(\mathbf{R}\eta) \frac{1 - z^2}{1 + z^2 - 2z\eta_3}.$$



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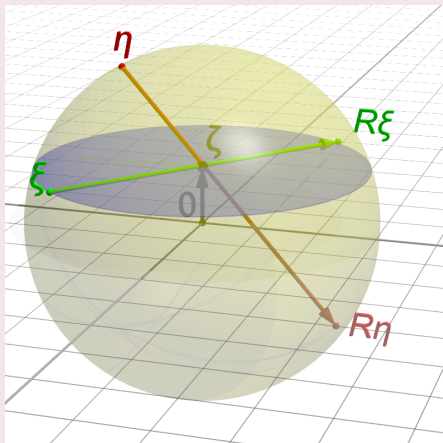
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Range of $\mathcal{U}$

Theorem

The generalized Radon transform

$$\mathcal{U}: \tilde{L}_e^2(\mathbb{S}^2) \rightarrow H_e^{1/2}(\mathbb{S}^2)$$

is continuous and bijective.

- ▶ $\tilde{L}_e^2(\mathbb{S}^2) = \left\{ f \in L^2(\mathbb{S}^2) \mid f(\boldsymbol{\eta}) = f(\mathbf{R}\boldsymbol{\eta}) \frac{1 - z^2}{1 + z^2 - 2z\eta_3} \right\}$
- ▶ $H_e^{1/2}(\mathbb{S}^2)$... Sobolev space of smoothness $1/2$ that contains only even functions

Sketch of proof

We show

$$\mathcal{U} = \mathcal{N}\mathcal{F}\mathcal{M}: \tilde{L}^2(\mathbb{S}^2) \rightarrow H_e^{1/2}(\mathbb{S}^2)$$

is continuous and bijective.

- ▶ $\mathcal{M}: \tilde{L}_e^2(\mathbb{S}^2) \rightarrow L_e^2(\mathbb{S}^2)$ is unitary
- ▶ $\mathcal{F}: L_e^2(\mathbb{S}^2) \rightarrow H_e^{1/2}(\mathbb{S}^2)$ is continuous and bijective
- ▶ $\mathcal{N}, \mathcal{N}^{-1}: L_e^2(\mathbb{S}^2) \rightarrow L_e^2(\mathbb{S}^2)$ are continuous and
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Inversion formula

Theorem

Let $z \in [0, 1)$ and $f \in \tilde{L}_e^2(\mathbb{S}^2)$. For $\eta \in \mathbb{S}^2$,

$$f(\eta) = \frac{1 - z^2}{2\pi(1 - zv)} \frac{d}{du} \int_0^u \int_{\mathcal{S}_z(\eta, w)} \mathcal{U}f \left(\frac{\left(\sqrt{1 - z^2 + \xi_3^2}(\xi_1, \xi_2), \xi_3 \right)}{\sqrt{1 - z^2 + z^2 \xi_3^2}} \right) ds(\xi) \cdot \frac{dw}{\sqrt{u^2 - w^2}} \Big|_{u=1},$$

where ds is the arc-length on the circle

$$\mathcal{S}_z(\eta, w) = \left\{ \xi \in \mathbb{S}^2 \mid \left\langle \xi, \left(\sqrt{1 + \eta_3^2 - 2z\eta_3(\eta_1, \eta_2)}, z - \eta_3 \right) \right\rangle = \frac{\sqrt{1 - w^2}}{z\eta_3 - 1} \right\}.$$

Inversion via Fourier expansion

- ▶ The last inversion formula is numerically unstable

- ▶ Use the factorization

$$\mathcal{U} = \mathcal{N}\mathcal{F}\mathcal{M}$$

- ▶ $\mathcal{M}^{-1}$ and $\mathcal{N}^{-1}$ can be computed explicitly

- ▶ For $\mathcal{F}^{-1}$: **Fourier expansion of the Funk–Radon transform** combined with the mollifier method (as regularization)

[Louis et al., 2011] [Hielscher & Q., 2015]

- ▶ Efficient implementation of the spherical Fourier transform available

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- ▶ For $\mathcal{F}^{-1}$: **Fourier expansion of the Funk–Radon transform** combined with the mollifier method (as regularization)

[Louis et al., 2011] [Hielscher & Q., 2015]

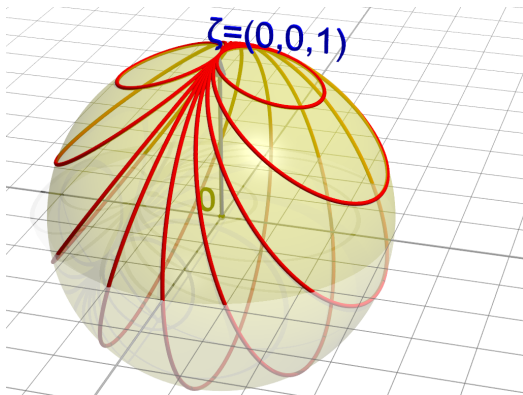
- ▶ Efficient implementation of the spherical Fourier transform available

[Keiner & Potts, 2008]

The case $z = 1$ Spherical Slice Transform

[Abouelaz & Daher, 1993]

$$\mathcal{V}f(\xi) = \int_{\langle \xi, \eta \rangle = \xi_3} f(\eta) \, ds(\eta)$$



- “Circles through the North pole”
- $\mathcal{V}$ is injective for all bounded functions

[Rubin, 2015]

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